

CS4084

Quantum Computing

Week 06 – Lecture 11-12

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Quick Review: What We have Covered Last Lecture I

- **Probabilistic States:** Compound systems associate probabilities with the **Cartesian product** of individual state sets.
- **System Independence:** Two systems are independent if $Pr((X, Y) = (a, b)) = Pr(X = a)Pr(Y = b)$.
- **Tensor Product:** The mechanism to combine vectors:
 $|\pi\rangle = |\phi\rangle \otimes |\psi\rangle$.
- **Vector Notation:** Follows lexicographical ordering, simplified as $|\phi\rangle|\psi\rangle$.
- **Linearity Property:** The tensor product is **bilinear**:

$$A \otimes (B + C) = A \otimes B + A \otimes C$$

Operations on probabilistic states

Example

A **controlled-NOT** operation on two bits X and Y:

If X = 1, then perform a NOT operation on Y, otherwise do nothing.

X is the **control bit** that determines whether or not a NOT operation is applied to the **target bit** Y.

Action on standard basis

$$|00\rangle \mapsto |00\rangle$$

$$|01\rangle \mapsto |01\rangle$$

$$|10\rangle \mapsto |11\rangle$$

$$|11\rangle \mapsto |10\rangle$$

Matrix representation

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Example..

Example

Here is a different operation on two bits (X, Y):

With probability 1/2, set Y to be equal to X, otherwise set X to be equal to Y.

The matrix representation of this operation is as follows:

$$\begin{pmatrix} 1 & 1/2 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1/2 & 1/2 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

Operations on probabilistic states

Definition

$$M = \sum_{a,b \in \Sigma} \alpha_{ab} |a\rangle\langle b| \quad \text{and} \quad N = \sum_{c,d \in \Gamma} \beta_{cd} |c\rangle\langle d|$$

$$M \otimes N = \sum_{a,b \in \Sigma} \sum_{c,d \in \Gamma} \alpha_{ab} \beta_{cd} |ac\rangle\langle bd|$$

An alternative, but equivalent, way to define $M \otimes N$ is that it is the unique matrix that satisfies the equation

$$(M \otimes N) |\phi \otimes \psi\rangle = M|\phi\rangle \otimes N|\psi\rangle$$

for every choice of vectors $|\phi\rangle$ and $|\psi\rangle$.

Example..

Example

$$\begin{pmatrix} \alpha_{00} & \alpha_{01} \\ \alpha_{10} & \alpha_{11} \end{pmatrix} \otimes \begin{pmatrix} \beta_{00} & \beta_{01} \\ \beta_{10} & \beta_{11} \end{pmatrix} \\ = \begin{pmatrix} \alpha_{00}\beta_{00} & \alpha_{00}\beta_{01} \\ \alpha_{00}\beta_{10} & \alpha_{00}\beta_{11} \end{pmatrix} \begin{pmatrix} \alpha_{01}\beta_{00} & \alpha_{01}\beta_{01} \\ \alpha_{01}\beta_{10} & \alpha_{01}\beta_{11} \end{pmatrix} \\ \begin{pmatrix} \alpha_{10}\beta_{00} & \alpha_{10}\beta_{01} \\ \alpha_{10}\beta_{10} & \alpha_{10}\beta_{11} \end{pmatrix} \begin{pmatrix} \alpha_{11}\beta_{00} & \alpha_{11}\beta_{01} \\ \alpha_{11}\beta_{10} & \alpha_{11}\beta_{11} \end{pmatrix}$$

Example..

Example

Recall this operation from Lesson 1:

$$\begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{pmatrix}$$

Suppose this operation is performed on a bit X , and a NOT operation is (independently) performed on a second bit Y .

The combined operation on the compound system (X, Y) then has this matrix representation:

$$\begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 1 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \end{pmatrix}$$

Quantum States

Quantum state vectors of multiple systems are represented by column vectors whose indices correspond to the Cartesian product of the individual systems' classical state sets.

Example

If X and Y are qubits, the classical state set for the pair (X, Y) is

$$\{0, 1\} \times \{0, 1\} = \{00, 01, 10, 11\}$$

These are examples of quantum state vectors of the pair (X, Y):

$$\begin{aligned} & \frac{1}{2}|0\rangle|0\rangle + \frac{i}{2}|0\rangle|1\rangle - \frac{1}{2}|1\rangle|0\rangle - \frac{i}{2}|1\rangle|1\rangle \\ & \frac{3}{5}|0\rangle|0\rangle - \frac{4}{5}|1\rangle|1\rangle \\ & |0\rangle|1\rangle \end{aligned}$$

Quantum States

Tensor products of quantum state vectors are also quantum state vectors.

Let $|\phi\rangle$ be a quantum state vector of a system X and let $|\psi\rangle$ be a quantum state vector of a system Y . The tensor product

$$|\phi\rangle \otimes |\psi\rangle$$

is then a quantum state vector of the system (X, Y) .

States of this form are called **product states**. They represent **independence** between the systems X and Y .

More generally, if $|\psi_1\rangle, \dots, |\psi_n\rangle$ are quantum state vectors of systems $X_1, \dots, X_n$, then

$$|\psi_1\rangle \otimes \dots \otimes |\psi_n\rangle$$

is a quantum state vector representing a product state of the compound system $(X_1, \dots, X_n)$.

Example..

Example

The quantum state vector

$$\frac{1}{2}|00\rangle + \frac{i}{2}|01\rangle - \frac{1}{2}|10\rangle - \frac{i}{2}|11\rangle$$

is an example of a product state:

$$\begin{aligned} \frac{1}{2}|00\rangle + \frac{i}{2}|01\rangle - \frac{1}{2}|10\rangle - \frac{i}{2}|11\rangle \\ = \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \right) \otimes \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle \right) \end{aligned}$$

The Bell Basis

Here are a couple of well-known examples of quantum state vectors for three-qubits.

GHZ state

$$\frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|111\rangle$$

W state

$$\frac{1}{\sqrt{3}}|001\rangle + \frac{1}{\sqrt{3}}|010\rangle + \frac{1}{\sqrt{3}}|100\rangle$$

Unitary Operations on Multiple Systems

Quantum operations on compound systems are represented by unitary matrices whose rows and columns correspond to the Cartesian product of the classical state sets of the individual systems.

Example

Suppose X has classical state set $\{1, 2, 3\}$ and Y has classical state set $\{0, 1\}$. This unitary matrix represents an operation on (X, Y) :

$$U = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{i}{2} & -\frac{1}{2} & 0 & 0 & -\frac{i}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{2} & -\frac{i}{2} & -\frac{1}{2} & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix}$$

Unitary Operations on Multiple Systems

The combined action of a collection of unitary operations applied independently to a collection of systems is represented by the *tensor product* of the unitary matrices.

That is, if $X_1, \dots, X_n$ are quantum systems, $U_1, \dots, U_n$ are unitary matrices representing operations on these systems, and the operations are performed independently on the systems, the combined action on $(X_1, \dots, X_n)$ is represented by the matrix

$$U_1 \otimes \dots \otimes U_n$$

In particular, if we perform a unitary operation U on a system X and do nothing to a system Y , the operation on (X, Y) we obtain is represented by the unitary matrix

$$U \otimes \mathbb{I}_Y$$

Not every unitary operation on a compound system can be expressed as a tensor product of unitary operations.

Example

Suppose that X and Y are systems that share the same classical state set Σ . The **swap operation** on the pair (X, Y) exchange the contents of the two systems:

$$\text{SWAP}|\phi \otimes \psi\rangle = |\psi \otimes \phi\rangle$$

It can be expressed using the Dirac notation as follows:

$$\text{SWAP} = \sum_{a, b \in \Sigma} |a\rangle\langle b| \otimes |b\rangle\langle a|$$

Example

The swap operation can be expressed using the Dirac notation as follows:

$$\text{SWAP} = \sum_{a,b \in \Sigma} |a\rangle\langle b| \otimes |b\rangle\langle a|$$

For instance, when X and Y are qubits, we find that

$$\text{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Example..

Example

$$\text{SWAP}|\phi^+\rangle = |\phi^+\rangle$$

$$\text{SWAP}|\phi^-\rangle = |\phi^-\rangle$$

$$\text{SWAP}|\psi^+\rangle = |\psi^+\rangle$$

$$\text{SWAP}|\psi^-\rangle = -|\psi^-\rangle$$

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

$$|\phi^-\rangle = \frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{2}}|11\rangle$$

$$|\psi^+\rangle = \frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle$$

$$|\psi^-\rangle = \frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|10\rangle$$

Example..

Suppose that X is a qubit and Y is an arbitrary system.

For every unitary operation U on Y , a **controlled- U** operation is a unitary operation on the pair (X, Y) defined as follows:

$$|0\rangle\langle 0| \otimes \mathbb{1}_Y + |1\rangle\langle 1| \otimes U = \begin{pmatrix} \mathbb{1}_Y & 0 \\ 0 & U \end{pmatrix}$$

Example

A controlled-NOT operation (where the first qubit is the control):

$$|0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes \sigma_x = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Example..

Example

A controlled- σ_z (or controlled-Z) operation:

$$|0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes \sigma_z = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

More Operations!

Example

A controlled-SWAP operation (on three qubits):

$$|0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes \text{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

This operation is also known as a **Fredkin operation** (or Fredkin gate).

More Operations!

Example

A controlled-controlled-NOT operation (on three qubits):

$$|0\rangle\langle 0| \otimes \mathbb{1} \otimes \mathbb{1} + |1\rangle\langle 1| \otimes (|0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes \sigma_x)$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

This operation is better known as a **Toffoli operation** (or Toffoli gate).

Summary: Key Takeaways I

- **Probabilistic Operations:** Transformations on compound systems use stochastic matrices acting on the Cartesian product of state spaces.
- **Matrix Tensor Products:** Joint operations are represented by the Kronecker product $(M \otimes N)$.
- **Quantum States:** Multiple qubits exist in the joint tensor product space $(\mathbb{C}^2)^{\otimes n}$.
- **State Vector Products:** Independent qubits combine as $|\phi\rangle \otimes |\psi\rangle$ following lexicographical ordering (e.g., $|00\rangle, |01\rangle, |10\rangle, |11\rangle$).
- **Unitary Multi-Systems:**
 - **Local:** $H \otimes I$ acts only on the first qubit.
 - **Global:** CNOT-like gates are inseparable and essential for entanglement.
- **Bilinearity:** Tensor products are linear in every "slot," allowing scalars to move: $\alpha(|A\rangle \otimes |B\rangle) = (\alpha|A\rangle) \otimes |B\rangle$.

Summary: Key Takeaways II

Next: Entanglement, Global Unitary Operations, and Multi-System Postulates.